



P14

Where Recursive Self-Improvement Converges

RSI 收敛：开源优化系统闭源优化流程

TwinsEarth · 智能体宇宙论文系列

Where Recursive Self-Improvement Converges

Working Title (EN): Where Recursive Self-Improvement Converges: Open-Source Optimizes the System, Closed Labs Optimize the Pipeline, and UDOS Makes Improvement Measurable — With a Structural-Organization View of Intelligence

Topical paper P14 • the sequel to P13 The Wall of Scaling Is a Paradigm Wall • mutually a theoretical base with flagship papers P0/P0b • UDOS Reasoning Engine v7.5.0 • 2026-09-20

In one sentence: the RSI (Recursive Self-Improvement) practices of seven major institutions in 2026 are converging to the same near-term form—**not the model rewriting its own weights, but the model optimizing "the way it obtains answers" (the Harness)**. The open-source camp optimizes the system, the closed-source camp optimizes the R&D pipeline, and UDOS does a third thing: making "becoming stronger" itself **measurable, falsifiable, auditable**.

Evidence scope (stated once for the whole paper): same as P13. UDOS numbers trace reports7/*.json, at three levels verified / cpu-proto / unverified; single seeds written as "pilots." External papers and industry events were verified online one by one on 2026-09-20: where the **existence of a paper/report** was verified, the arXiv number or official page is given; the efficiency multiples, penetration rates, and concurrency within are all **paper reports or company disclosures, not independently verified**; entries with no evidence are explicitly removed. This paper **does not claim any existing system has consciousness, a self, or complete RSI**.

Abstract

Recursive Self-Improvement (RSI) is turning from a science-fiction narrative into an auditable engineering object. This paper does three things. First, it reviews seven representative 2026 RSI routes and verifies the status of their materials: on the open-source side, DeepSeek Harness/Cordis (MIT license, developer preview released 2026-08, kernel derived from the Koishi ecosystem running four years with over four thousand plugins; companion paper A Programming Paradigm for Spatiotemporal Composability), Kimi's Attention Residuals (arXiv:2603.15031, paper reports about $1.25\times$ compute efficiency), the Infra Agent driven by Zhipu GLM-5.3 (company discloses raising end-to-end throughput about $3\times$ in under two weeks on over 100,000 domestic chips while explicitly self-stating "not yet complete RSI"), Meta's AIRA (arXiv:2605.15871) and the MetaRSI/RSI² from CosmosMind et al. (arXiv:2609.06396, three-operator unification, paper reports small models +10.9 points on average); on the closed-source side, OpenAI's Research acceleration: The view inside OpenAI (2026-09-06, announcing the "automated research intern" milestone, targeting the 2028-03 "automated researcher"), Google/DeepMind et al.'s Dream-RSI (arXiv:2609.14858, discovery-tree replay, paper reports up to $162\times$ fewer agent calls), and Anthropic's R&D Automation Index (official pages disclose Claude already "leading" about 26% of its own R&D, over 90% reaching collaboration level, about 30,000 R&D agents concurrent, while publicly calling for prudence).

Second, we give convergence criteria: the optimization targets of these works move up the same chain—prompt → structured context → workflow → harness code → optimizer code—and all 2026 production cases fall in the first four segments, while the fifth (autonomously changing weights/optimizer) has no controlled public evidence. The open-source camp optimizes the **system** (Harness, architecture, Infra), the closed-source camp optimizes the **pipeline** (R&D automation, exploration policy, penetration

monitoring), corresponding on P0b's three-section curve respectively to section 1 (Harness-RSI) and the early part of section 2 (Model-RSI).

Third, the theoretical part, we connect this engineering convergence to a more fundamental proposition: **life originates in structural organization, and intelligence is also a kind of structural organization; carbon is not the privilege of intelligence—connectivity is**. Hebb's rule and gradient descent are the same principle (encoding experience as connections) implemented on two substrates; learning shapes structure and structure in turn constrains subsequent learning, a bidirectional causality shared by carbon and silicon. The ultimate form of RSI can therefore be precisely defined as **structural self-modification**—the system begins to modify "the structure that determines subsequent modifications." UDOS's unique position is not joining in building stronger systems but providing the measurement instrument for structural self-modification: BFT-lite stop decisions, TransferBundle handoff contracts, internal-market settlement conservation, and evidence grading constitute the engineering base for "knowing where one does not know and where one is not trustworthy." This paper gives four falsifiable conditions, five hard predictions, and a consciousness-boundary statement.

Keywords: recursive self-improvement; Harness; multi-agent governance; structural organization; scaling law; evidence grading; falsifiability; AI safety

Structured abstract (background problem → method → evidence/results → contributions)

- **Background problem:** in the RSI narrative marketing and evidence are mixed, the strong claim "AI rewrites its own brain" is severely disconnected from actual engineering progress, and a unified comparison framework and credibility grading are missing.
- **Method:** (1) event-level verification of the seven routes (does the paper exist, is the number a one-sided claim or independently reproduced); (2) use the "optimization-target chain" to locate each route's real landing point; (3) map the convergence position with P0b's three-section curve; (4) give the engineering definition of RSI and falsifiable conditions with the structural-organization view.
- **Evidence/results:** the existence of 7 classes of papers/reports verified; all efficiency/penetration multiples are one-sided claims; the fifth segment (autonomously changing weights) has zero controlled evidence; UDOS governance mechanisms have verified test guards (BFT 24/24, conservation true, 26 contract tests).
- **Contributions:** the convergence map of two camps × three sections; the five-segment optimization-target chain; the precise definition of "structural self-modification"; rewriting the philosophical proposition "intelligence is structural organization" into four falsifiable conditions.

1 Introduction: first tighten the term RSI

"Recursive self-improvement" is the most easily abused. OpenAI gave a restrained working definition in its 2026-09 internal report: if an agent only helps write code, debug infrastructure, and analyze experimental results, that is closer to "research acceleration"; only when AI can continuously participate in **topic selection, solution design, experiment execution, result judgment, and the next round of**

model improvement and form a stable feedback loop is it closer to RSI (official report, page verified to exist, phrasing a paraphrase). This paper adopts this tightening and distinguishes three layers:

- **Task-layer self-improvement:** improving the policy for completing tasks under a fixed model and fixed tools (e.g. exploration policy, prompts, workflows).
- **System-layer self-improvement (Harness-RSI):** improving the runtime system outside the model —tools, sandboxes, plugins, context management, Infra.
- **Model-layer self-improvement (Model-RSI):** modifying model parameters or architecture under bounded, verifiable conditions; its meta-level (improving "the improvement system itself") is called Meta-RSI.

All public production evidence in 2026 concentrates in the first two layers; the third has only candidate prototypes and paper reports, with no controlled public evidence of "AI autonomously rewriting its own weights and stably benefiting." This distinction is the baseline of the whole paper.

2 Theoretical framework: intelligence is a kind of structural organization

Before entering vendor comparison, a framework that can accommodate both carbon and silicon is needed, otherwise the RSI discussion degenerates into a parameter race.

2.1 Learning is establishing connections

Figure 3 Structural-organization isomorphism of carbon and silicon

Figure 3 Hebbian learning and gradient descent differ in substrate but are isomorphic in the connection principle; both sides satisfy the bidirectional causality "learning shapes structure, structure constrains learning."

Neuroscience's Hebb's rule says "neurons that fire together wire together": skills and memory are not stored in some neuron but distributedly encoded in the weight patterns of synaptic connections. Large-model training is mathematically the backpropagation version of the same principle: gradient descent adjusts the weight matrix, attention lets each token dynamically decide "whom to attend to" by context, and MoE routing lets experts connect dynamically by task. Kimi's Attention Residuals also changes the **allocation** of connections—letting each layer selectively aggregate historical layers rather than accumulate indiscriminately (arXiv:2603.15031, mechanism description per its abstract).

P0b's central formula therefore gains a cognitive meaning:

$$\text{Capability} \approx F(S; N, D, C), \quad S = (T, G, M) \tag{1}$$

The human brain's about 86 billion neurons become smart not by "many parameters" but by **connection structures** such as cortical folding, white-matter wiring, and modular division; model capability is likewise first a function of structure and only secondarily a function of scale.

2.2 Structure shapes learning and also constrains learning

This is a symmetric bidirectional causality. On the human side: music training changes auditory-cortex gray matter and corpus-callosum white matter, and the changed structure both makes subsequent music learning easier and makes it harder to segment sound in non-musical ways; after the language sensitive period, Broca's/Wernicke's-area connections solidify, the structural reason adults find new languages harder. On the machine side: attention heads differentiate during pretraining into syntax heads, coreference heads, fact-retrieval heads, and once the structure solidifies, SFT/RLHF can only fine-

tune on the established wiring; if MoE routing collapses to a few experts, subsequent training can hardly restore diversity.

This is precisely the cognitive version of P0b's "structure determines the form of the scaling function": **how much data the current structure can absorb and into what shape it absorbs it determines the sign (positive/zero/negative) of the next step's returns.**

2.3 Carbon is not the privilege, connectivity is

Origin-of-life research understands the earliest life as an **autocatalytic network**—a set of molecules mutually catalyzing to form a closed loop and maintain its own structural organization; the key is the network structure (who catalyzes whom), and carbon is only one chemical substrate. Intelligence likewise: P0b sees isomorphic three-section curves on seven substrates—chemical bonds, proteins, cells, GPUs, battery cells, neurons, text vectors—suggesting intelligence depends on the "structure + scale" organization rather than a specific substrate. The real differences between carbon and silicon lie in speed, plasticity, energy, constraint types, and learning modes (online/offline, rollback-able or not), not in "whether intelligence is possible."

The **structural-prior** thread brings this into engineering: BPE is an input structural prior, attention is a sequence structural prior, and UDOS's layered topology (communication reduced from $\$O(N)\$$ to $\$O(\log N)\$$) is an orchestration structural prior. Intelligence is not learned from scratch but emerges under structural-prior constraints; silicon's uniqueness is that structural priors **can be designed, searched, self-modified**—Meta AIRA letting agents autonomously search architectures is searching the structural prior itself (arXiv:2605.15871, verified to exist).

2.4 The engineering definition of RSI: structural self-modification

From this, the ultimate form of RSI can be precisely defined: **structural self-modification—the system modifies "the structure that determines subsequent modifications."** Changing weights moves parameters under the same structure (walking along the curve); changing the Harness, architecture, exploration policy, or optimizer changes the structure itself (switching curves). The Theseus-Labs-style RSI grading level "improving the improvement system itself" [CITATION NEEDED: grading-framework original pending verification] and the three-operator unification MetaRSI claims (Data-/Harness-/Model-RSI sharing a Loop Kernel, arXiv:2609.06396, verified to exist) are two candidate implementations of this definition. P0b's F5 prediction also gives the criterion for whether it is worth continuing: if the cross-substrate master curve can collapse, structural self-modification is predictable, guidable, auditable; if not, it is just "each has its own scaling."

3 The optimization-target chain and the RSI lineage

Figure 1 The optimization-target progression chain

Figure 1 The landing point of recent RSI is letting the model optimize "the way answers are obtained." All 2026 production cases concentrate in the first four segments; the fifth (autonomously changing optimizer/weights) has no controlled public evidence. The chain statement synthesizes the OpenAI report and industry reviews [CITATION NEEDED].

We arrange "what AI is allowed to change" into five levels: **prompt → structured context → workflow → harness code → optimizer code**. The farther right, the "deeper" the self-improvement, the higher the rollback cost, and the harsher the verifier requirements. Paired with the AL0–AL5 automation grading Anthropic cites from Epoch AI (official page, AL4 "leading" means engineers give only high-level

goals and AI completes most of the end-to-end work), any RSI claim can be located: which segment of the chain it changes to, which AI level it automates to, and whether there is a verifiable judge.

4 The open-source camp: optimizing the "system"

4.1 DeepSeek Harness / Cordis: making the self-evolution environment into revocable plugins

DeepSeek Harness (dsh, MIT license, v0.1 developer preview released 2026-08-13, repo deepseek-ai/deepseek-harness; multi-source verification consistent) proposes "Agent = Model + Harness" and makes model adapters, tool registries, session logs, and even the agent loop itself into plugins. Its kernel Cordis comes from the Koishi ecosystem (reported to have run about four years with over four thousand community plugins), achieving "spatiotemporal composability" through dependency injection and reversible side effects: in time, after a plugin is unloaded its registered services, events, and side effects can be traced and revoked; in space, plugins declare dependencies and the system automatically re-coordinates. Companion paper A Programming Paradigm for Spatiotemporal Composability (DeepSeek and Peking University, GitHub cordiverse/paper). **Positioning: the runtime base of Harness-RSI.** The "GitHub 22,600+ stars" number was not verified from an official page this time, marked [CITATION NEEDED], not written into conclusions.

4.2 Kimi Attention Residuals: raising the self-improvement ceiling from the architecture base

arXiv:2603.15031 (2026-03, verified to exist) points out fixed unit-weight residual accumulation causes hidden-state magnitude to grow with depth and early-layer contributions to be diluted; AttnRes replaces fixed accumulation with softmax attention over prior-layer outputs, and Block AttnRes reduces memory/communication complexity from $\mathcal{O}(Ld)$ to $\mathcal{O}(Nd)$. The paper reports on Kimi Linear (48B total/3B activated) obtaining about $1.25\times$ compute efficiency with 1.4T-token pretraining, inference latency increase $<2\%$ (**paper report, not independently verified**). **Positioning: an architecture fix paving the way for Model-RSI.**

4.3 Zhipu GLM-5.3 Infra Agent: the first domestic case to run into a production environment

Per Zhipu's technical blog and Tang Jie's 2026-09-17 long post (multi-source verification by Yicai, 36Kr, InfoQ, Securities Times, et al.): the GLM-5.3-driven Infra Agent on over 100,000 domestic accelerators participated from scratch in building the production inference service for GLM-5.3-Flash, raising end-to-end throughput to about $3\times$ the initial baseline in under two weeks (Tang Jie personally states $3.2\times$), located the Python GIL concurrency blockage in KV Transfer and pressed the over-20% loss of Prefill+KV Transfer to about 1%, with the KDA Decode operator reporting $1.71\times$. **Most critical is its honest boundary:** the team explicitly states goal setting, boundary division, and risk judgment remain with human engineers, "not yet complete RSI," and what is closed is only the first puzzle piece of "objects that can be improved." **Positioning: production evidence of Harness/Infra-layer RSI + restrained self-characterization.**

4.4 Meta AIRA and CosmosMind MetaRSI: architecture search and meta-recursion

- **AIRA** (FAIR/Meta, arXiv:2605.15871, 2026-05, verified to exist): AIRA-Compose uses 11 agents within a 24-hour compute budget to compose Attention/MLP/Mamba primitives, AIRA-Design lets agents

directly write code to implement new attention; the paper reports finding 14 new architectures, AIRAformer-D / AIRAhybrid-D at 1B scale respectively 2.4/3.8 percentage points above Llama 3.2, some variant about 54% faster (**paper report, not independently verified**).

- **MetaRSI/RSI²** (arXiv:2609.06396, 2026-09, CosmosMind et al. over ten institutions, verified to exist): Data-RSI synthesizes validated training records, Harness-RSI changes scaffolding at zero training cost with five-slot typed patches, Model-RSI changes parameters under bounded recipes, the three sharing one Loop Kernel with a dual-axis optimizer deciding operator order; the paper reports 3B-activated small models +10.9 points on average, flagship models +7.3 points on average (**paper report, not independently verified**). Concurrent works such as arXiv:2609.14857 (ModularRSI) show the direction is rapidly differentiating.

The common feature of the open-source camp: improvement happens outside the model or at the architecture base, **not directly changing online weights**, rollback-able, composable, open-source auditable.

5 The closed-source camp: optimizing the "pipeline"

5.1 OpenAI: automated research intern → automated researcher by 2028

Research acceleration: The view inside OpenAI (2026-09-06, official page verified) discloses the penetration of coding agents into internal R&D, announces reaching the "automated AI research intern" milestone set last autumn, with the next target the 2028-03 "automated AI researcher"; media paraphrase internal metrics such as "each person-day corresponds to 3.1 agent workdays" (**media number, not independently verified**) and note 4–8-hour tasks still need at least one human intervention. The report also stresses RSI must be synchronized with democratic governance. **Positioning: a roadmap for R&D-pipeline automation, not a declaration that RSI is complete.**

5.2 Google/DeepMind: Dream-RSI—improving exploration policy in a "dream"

Dream-RSI (exact title Recursive Self-Improvement through Evolving Worlds, arXiv:2609.14858, 2026-09-14, Google/DeepMind/UMD/UVA, verified to exist) keeps the underlying coding agent unchanged and only makes exploration itself explicit and programmable: one online exploration forms a discovery tree, and subsequent policies directly "dream" and replay on historical trees, obtaining near-zero-cost offline policy evaluation. The paper reports competitive quality on three task types—Lasso solvers, mathematical optimization, GPU kernel engineering—up to $162\times$ fewer agent calls than SimpleTES and about $1.7\times$ over the fixed baseline, and observes human guidance is instead worse (**all paper reports**). Both authors and third parties are clear: model, evaluator, tools, and policy architecture stay fixed, and it is a recursive improvement at the **exploration-policy layer**, not complete model self-modification.

5.3 Anthropic: the most aggressive internal data + the most prudent public posture

Two official Anthropic Institute pages (Measurements for understanding the pace of AI development inside frontier labs and When AI builds itself, 2026-09, verified) disclose: building an R&D Automation Index (adopting Epoch AI's AL0–AL5); as of 2026-08, Claude reached AL4 "leading" in about 26% of its own AI R&D, over 90% at least AL3 collaboration level (that proportion was under 1% in 2026-02); on the most-used internal platform about 30,000 R&D/engineering agents concurrently daily; in August over 1 billion decisions were tracked, monitoring triggering about 0.002% blocks and about 20,000 out-of-bounds operations paused (**all company-self-reported scope**). Co-founder Jack Clark's statement

about "over 60% probability of human-free RSI by end-2028 and calling to slow down" appears this time only in secondary reports, marked [CITATION NEEDED]. **Positioning: making the RSI penetration rate a monitorable metric while institutionalizing the safety brake.**

The common feature of the closed-source camp: improvement happens at the **R&D-pipeline** layer—AI participates in developing the next generation of AI, but topic selection, boundaries, and risk judgment remain on the human side.

6 UDOS's position: optimizing "the process of becoming stronger itself"

Figure 2 The three camps' positions on the three-section curve

Figure 2 The 2026 open/closed-source cases concentrate in section 1 (Harness-RSI) and probe into section 2 (Model-RSI); UDOS provides no new capability point but the coordinate system and measurement instrument for the whole curve.

Both open and closed source answer "how to make AI stronger." UDOS answers a different question: **how can "becoming stronger" itself be measurable, falsifiable, auditable?** This corresponds to P12's evidence grading and the engineering of Confucius's "to know what one knows and what one does not know" (P0c). Six instruments each guard one baseline:

Instrument	Guards what	Controlled result (reports7)	Evidence level
Layered hybrid topology (P1)	Communication not exploding quadratically with headcount	Million-node center load 20.97M→9, rounds 8.39M→14	cpu-proto extrapolation, counts verified
BFT-lite stop decisions (P2)	Single-point/Byzantine members cannot decide "done"	≥ 1 , at $n=7, f=2$ 24/24 correct	10 property tests verified
Quality saturation law (P9)	Homogeneous expansion not producing false gains	Equal-weight bad expert 0.2131 vs calibrated weighting 0.0092 (about $23\times$)	≤ 2 verified, > 2 cpu-proto
TransferBundle (P10)	Handoff not losing context, not transferring responsibility	Six-field mechanical validation	26 contract tests verified; failure rate [RESULT NEEDED]
Internal-market conservation (P11)	The accounts of a million agents balancing	balance=paid-slashed, 24/16/1 conservation true	10 unit tests verified
Evidence grading (P12)	Numbers not drifting, prototypes not written as measurements	Three-tier labels + baseline JSON regression gate	Engineering archive across 30+ versions

This instrument set is fully consistent with the structural-organization view: since intelligence is a function of structure, whether "improvement" happens must be measurable on **structural quantities** (topology fan-in, independent-source count, contract completeness, conservation residual), not just by listening to the model self-report "I am stronger." P0c's triple (M_{self}, U, A) —what I know, what I don't know, how to act accordingly—is precisely the RSI safety loop: each time structural self-modification happens, the system must be able to recalibrate its own knowledge boundary.

7 Convergence judgment: where the three meet

Placing the seven routes and the UDOS framework side by side yields three levels of convergence:

1. **Object convergence:** recent RSI = optimizing the way answers are obtained (Harness), not rewriting the brain. Production evidence on the five-segment chain stops at the fourth segment.
2. **Curve convergence:** open-source optimizing systems and closed-source optimizing pipelines are both structural phase-change actions on P0b's three-section curve; the scale pressure of Zhipu's 100,000 cards and Anthropic's 30,000 agents is precisely the engineering signal of the section 1→section 2 phase change.
3. **Governance convergence:** the larger the scale, the more "measurable, rollback-able, auditable" is not optional—DeepSeek uses reversible side effects for plugin rollback, OpenAI stresses governance synchronization, Anthropic uses block rate as a brake, and UDOS uses Owner/Trace/Stop Condition/conservation as the formalized baseline.

In one sentence: **open source optimizes the system, closed source optimizes the pipeline, UDOS optimizes "the measurement and governance of structural self-modification"; all three ultimately converge onto the same three-section curve, only standing at different positions.**

8 Falsifiable conditions, hard predictions, and the consciousness boundary

8.1 Four falsifiable conditions of "intelligence is structural organization"

- C1: if some intelligent capability is found unrelated to connection structure, determined only by substrate material (e.g. "pure quantum intelligence" not relying on any connection topology), then "intelligence is in connections" is overturned—no evidence currently.
- C2: if the same connection structure on carbon/silicon shows an **incommensurable difference in capability kind** (rather than efficiency difference), then "carbon is not privileged" must weaken to "carbon has unique capabilities"—the differences currently observed run along task distributions, not substrate kinds.
- C3: if this view derives "any structure can reach any intelligence," it is wrong; P0b already limited "adding scale under the wrong structure gives negative returns," structure determines the reachable set and does not guarantee reaching.
- C4: if this view erases the role of dynamics (learning rules, signal propagation), it is one-sided; structure is the static skeleton, dynamics the dynamic process, and intelligence is the joint product of both (P0 already distinguished structure \$\$\$ from dynamics $\mathrm{D}_{\{\mathrm{dyn}\}}$).

8.2 Five hard predictions

- F1: within 2027 mainstream "RSI breakthroughs" will still occur in the first four chain segments; if a controlled, reproducible, negative-result-bearing public report of the fifth segment (autonomously changing optimizer/weights) appears, this paper's convergence timeline must be rewritten.
- F2: Harness-RSI returns obey P0b's three sections: after plugin/agent count exceeds the true independent-source count, marginal returns flatten or turn negative (isomorphic to P9); a plugin-count vs task-success-rate regression can be run on the Cordis/Koishi ecosystem.
- F3: Dream-RSI-style replay returns are monotonically non-decreasing with discovery-tree size but decay across task families; zero cross-family transfer downgrades it to within-task caching.

- F4: R&D automation penetration (AL4 share) and block/rollback event rates will rise together; if only penetration rises while brake metrics are not public, the penetration number is not credible.
- F5: if the cross-substrate master curve fails to collapse on ≥ 3 non-language substrates (chips, battery cells, agent topology), the strong claim "structural self-modification is uniformly predictable" is falsified.

8.3 Consciousness-boundary statement (a separate section)

This paper strictly distinguishes **engineering self-modification** from **consciousness/self**. All "self-improvement," "dreaming," and "leading" in the whole paper refer to observable task/system/pipeline behavior, not claiming any 2026 system has phenomenal consciousness, a self-model, or autonomous goals, nor endorsing that consciousness theories such as Global Workspace Theory (GWT) are already implemented on machines. The "knowing one doesn't know" P0c handles is a calibratable engineering quantity (coverage, selective refusal, identifiability), not subjective experience. RSI safety discussions should be built on measurable structural quantities and rollback mechanisms, not the anthropomorphic judgment of "whether it wakes up."

9 Counter-arguments

1. "The two-camp division is post-hoc labeling." Partially holds—individual works cross categories (e.g. AIRA both changes architecture and automates the pipeline); but the five-segment-chain locating method is still uniquely decidable for a single work, and classification serves comparison rather than essentialization.
 2. "All multiples are self-reports, the conclusions are empty." This paper's core conclusions (convergence on the Harness, no evidence in the fifth segment, governance as the baseline) **do not depend on any single multiple**; all multiples can be falsified without shaking the main conclusion.
 3. "UDOS is only a CPU toy; extrapolating to 30,000 agents is baseless." The extrapolation risk is agreed, so topology extrapolation is marked cpu-proto and governance tests verified, claiming only mechanisms hold not constants; F4/F5 are dedicated extrapolation tests.
 4. "The structural-organization view is not falsifiable, it is philosophical rhetoric." Section 8.1 gives four explicit kill conditions; only if none of the four can be triggered by any evidence should the framework be discarded as metaphysics.
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10 Conclusion

The 2026 RSI evidence points to a calm and clear judgment: the short-term feasible self-improvement is not AI rewriting its own brain but AI optimizing the way it obtains answers. The open-source camp makes this into revocable, composable systems, the closed-source camp into monitorable, milestoneable R&D pipelines, and neither has yet crossed the "autonomously changing weights" line. Life and intelligence both originate in structural organization, carbon is not the privilege—connectivity is; therefore the engineering essence of RSI is structural self-modification, and once structural self-modification happens it must be matched by structure-level measurement and governance—knowing what changed, whether it can roll back, whether the accounts conserve, and where the boundaries are. UDOS's six instruments do not produce intelligence; they produce **trustworthy statements about intelligence**. On the eve of RSI, this may be closer to the core of the problem than one more clever agent.

References (verification status marked per entry; external entries all verified online 2026-09-20)

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3. Anthropic Institute. Measurements for understanding the pace of AI development inside frontier labs and When AI builds itself. Official pages, 2026-09 (**verified**; 26% AL4, >90% AL3, about 30k concurrency, 0.002%/about 20k blocks are company self-reports; AL0-AL5 cited from Epoch AI; Jack Clark 60%/slow-down call is a secondary report [CITATION NEEDED]).
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7. Agentic Discovery of Neural Architectures: AIRA-Compose and AIRA-Design. arXiv:2605.15871, 2026-05 (**verified**; 11 agents/24h budget in the abstract, 14 architectures/ $+2.4/+3.8/54\%$ are paper reports).
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9. The Economics of Recursive Self-Improvement. arXiv:2609.15802, 2026-09 (**verified to exist**, feedback-loop elasticity-product modeling, not cited in depth).
10. UDOS this volume: P0 Structure Is the Intelligence, P0b Structure + Scale Is the Intelligence, P0c Metacognition, P1/P2/P9/P10/P11/P12; numbers trace `reports7/*.json`.
11. Theseus Labs RSI grading, the "prompt $\rightarrow$... $\rightarrow$ optimizer code" chain: industry-review scope, original pending [CITATION NEEDED].

Appendix A Reproduction commands and test index

- BFT-lite: `reports7/bft_lite*.json` ($\$n=7, f=2\$, 24$ work orders, 10 property tests).
- Conservation: `reports7/internal_market*.json` (budget 24/paid 16/slashed 1, 10 unit tests).
- Contracts: TransferBundle six fields, 26 contract tests (see P10's appendix).
- Quality saturation: `reports7/expert_mixture*.json` ($\$a=0.009252, b=-4.233\times 10^{-5}\$, equal-weight bad expert 0.2131).$
- Topology: `reports7/topology_scale*.json` (actual nodes 73/585/4681/37449/299593/2396745; center load 640...20971520 vs layered constantly 9).

Appendix B Evidence ledger

- **verified:** BFT 24/24, conservation true, 26 contract fields, 10 market units, topology-count programs.
- **cpu-proto:** million-level extrapolation (20.97M→9, 8.39M→14), quality saturation \$k>2\$ segment, throughput hump.
- **unverified (one-sided claims):** 1.25×, 162×, 1.7×, about 3×/3.2×, 20%→1%, 1.71×, +10.9, +7.3, +2.4/+3.8/54%, 26%, >90%, 30k concurrency, 0.002%/20k, 3.1 agent workdays, over four thousand plugins, 2028-03 milestone.
- **No evidence/not verified:** 22,600 stars, Jack Clark's 60% statement, Theseus grading original.

Appendix C Internal-review record (five-dimension self-assessment before submission, read-only)

- Novelty: high (the first engineering comparison unifying seven RSI routes by the five-segment chain + three-section curve with a verification ledger).
- Evidence strength: external events strong, multiples weak; internal mechanisms strong, extrapolation weak.
- Main gaps: independent reproduction of external multiples; ≥ 30 seeds; formal proof of the RSI chain; an item-by-item mapping table to Epoch AI AL grading.
- Conclusion: **working-paper status**; after completing F1–F5 it can target AI governance/safety workshops or TMLR stance + empirical mixed tracks [CITATION NEEDED: venue selection checked separately].

Author's intended statement

Fang Wenxin (intended). UDOS Reasoning Engine v7.5.0 paper volume • topical paper P14. This paper's citations of external companies and products are only for academic commentary, trademarks belonging to their respective owners; all external numbers are marked "paper report/company disclosure/not verified" and do not constitute investment or procurement advice. This paper does not claim any existing AI system has consciousness.